

DEEP LEARNING — DL26

Knowledge Distillation for Mobile Action Recognition

Compressing a 3D ResNet-50 into an ultra-lightweight MobileNet3D on UCF-101 via logit-based, attention transfer, and cross-frame distillation.

STUDENT

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TRACK

Track 6 — KD for Mobile Action Recognition

INSTITUTION

Università degli Studi di Catania — DMI

ADVISOR

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REPOSITORY

github.com/danii909/KD_Project

The Capacity Gap

Goal: Compress a 3D ResNet-50 teacher (pretrained on Kinetics-400) into a MobileNet3D student for on-device video action recognition on UCF-101 (101 classes).

TEACHER

3D ResNet-50

88.82%

TEST ACCURACY

~128 MB

6.21 ms GPU

76.6 ms CPU

ACCURACY GAP

-29.3 pp

SIZE REDUCTION

13.5×

INFERENCE SPEED

1.82×

1.66× CPU (BS=1)

STUDENT

MobileNet3D

59.48%

TEST ACCURACY

~9.5 MB

3.42 ms GPU

46.1 ms CPU

UCF-101 & Group-Aware Split

101
ACTION CLASSES

13,320
VIDEO CLIPS

24f
@ 112×112 PX

Original Split Structure

UCF-101 is officially divided only into Train and Test partitions, **lacking an actual validation set** required for proper model selection.

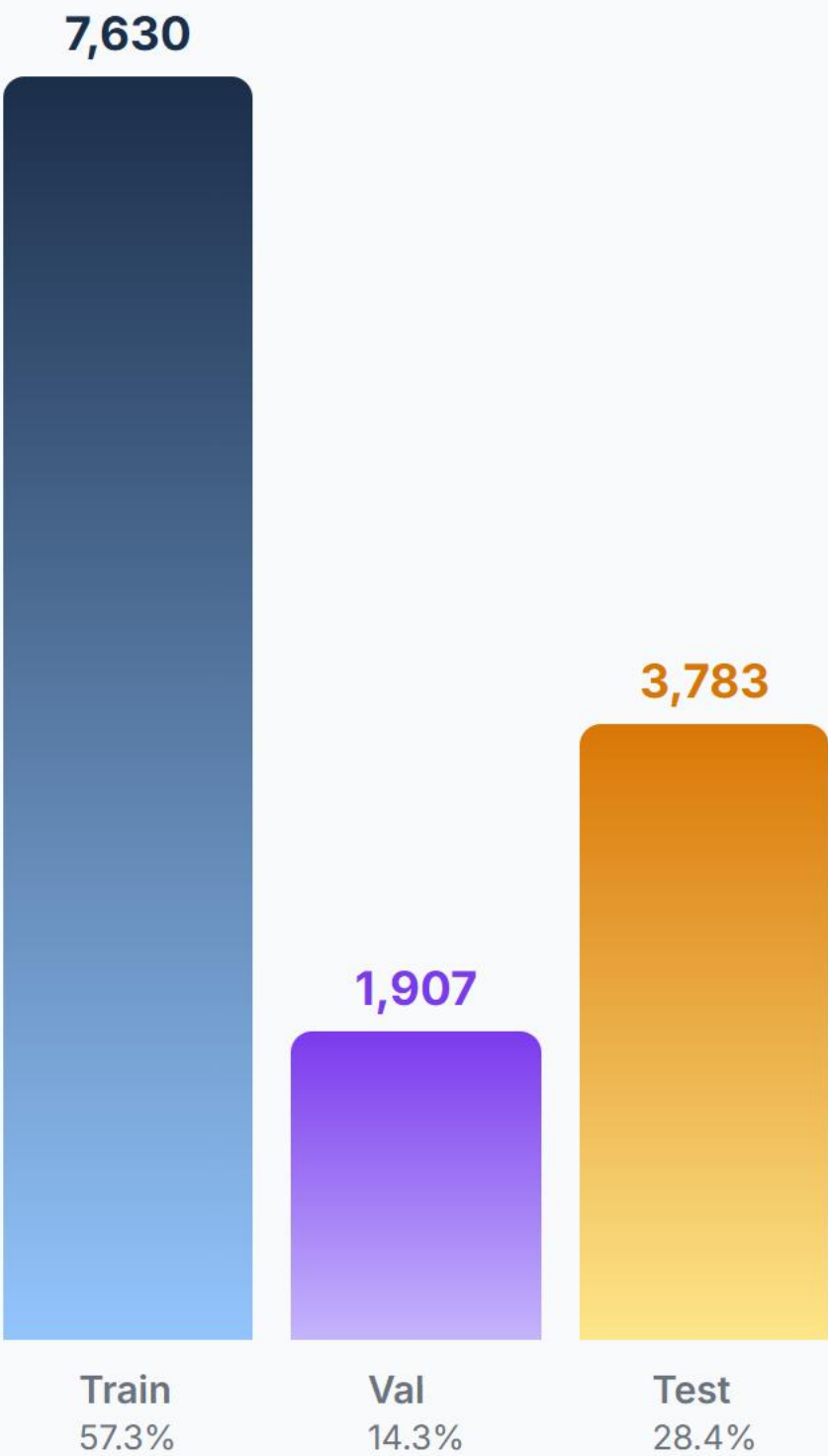
⚠ Data Leakage Risk

Correlated clips from the same source video share actor & background. Random splits cause **spatial leakage** and inflated accuracy.

✅ Group-Aware Split

Isolation by group ID (**_gXX_**). No source video shared across Train / Val / Test.

NEW SPLIT DISTRIBUTION



Knowledge Distillation Loss

Balanced objective function for knowledge transfer

$$\mathcal{L}_{\text{total}} = \underbrace{\alpha \cdot T^2 \cdot D_{\text{KL}} \left(\sigma(z^S / T) \parallel \sigma(z^T / T) \right)}_{\mathcal{L}_{\text{soft}}} + \underbrace{(1 - \alpha) \cdot \mathcal{L}_{\text{CE}}(z^S, y)}_{\mathcal{L}_{\text{hard}}}$$

$\alpha = 0.7$

TARGET BALANCING

Always set to **0.7** for the majority of experiments to prioritize mimicking the Teacher's soft targets (70%) over the hard ground truth labels (30%).

$T \in \{1, 5, 8, 10, 20\}$

LOGIT SCALING

The main hyperparameter varied across experiments to control the smoothness of probability distributions, extracting latent inter-class similarities.

Augmentation Trade-off & Data Leakage

All tests: T=10, a=0.7, 24f · Why light augmentation was chosen

1. Minimal Augmentation

WITH LEAK

65.7%

Top-5: 87.8%

NO LEAK

59.3%

Top-5: 83.6%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Disabled (0.0)
- **Random Erasing:** Disabled
- **Temporal stride:** Fixed (max_stride=1)

2. Light Augmentation ✓

WITH LEAK

69.5%

Top-5: 90.3%

NO LEAK

64.5%

Top-5: 87.4%

CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Light (0.1)
- **Random Erasing:** 5% probability
- **Temporal stride:** Fixed (max_stride=1)

3. Strong Augmentation

WITH LEAK

55.83%

Top-5: 83.27%

NO LEAK

50.4%

Top-5: 78.5%

CONFIGURATION

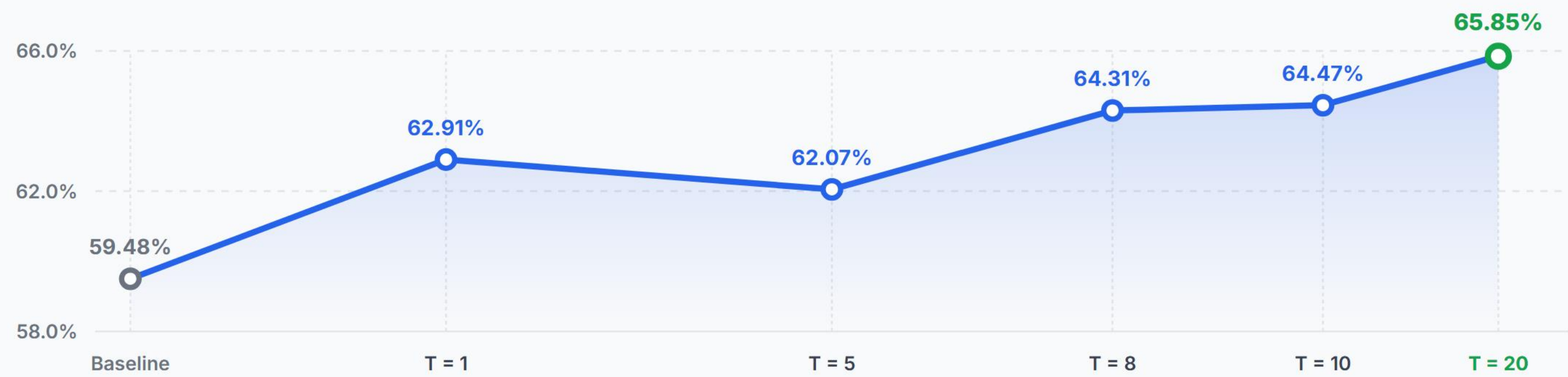
- **Spatial crop:** Resized Crop (0.6—1.0)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Strong (0.2)
- **Random Erasing:** 15% probability
- **Temporal stride:** Variable (max_stride=2)

Temperature Sweep & Main Result

Systematic analysis of Student behavior as T varies · a = 0.7 fixed

TEMPERATURE	BEST VAL ACC	EPOCH	TRAIN ACC	TEST TOP-1	TEST TOP-5	Δ VS BASELINE
Minimal Augmentation (no KD)	59.18%	—	99.81%	59.48%	82.77%	ref.
T = 1	62.75%	60	98.44%	62.91%	85.59%	+3.43%
T = 5	63.12%	57	95.87%	62.07%	87.02%	+2.59%
T = 8	64.57%	56	97.59%	64.31%	87.68%	+4.83%
T = 10	64.85%	60	98.28%	64.47%	87.44%	+4.99%
T = 20	65.18%	59	99.05%	65.85%	87.81%	+6.37%

TEST ACCURACY VS. TEMPERATURE TREND



Latent Space Analysis via t-SNE

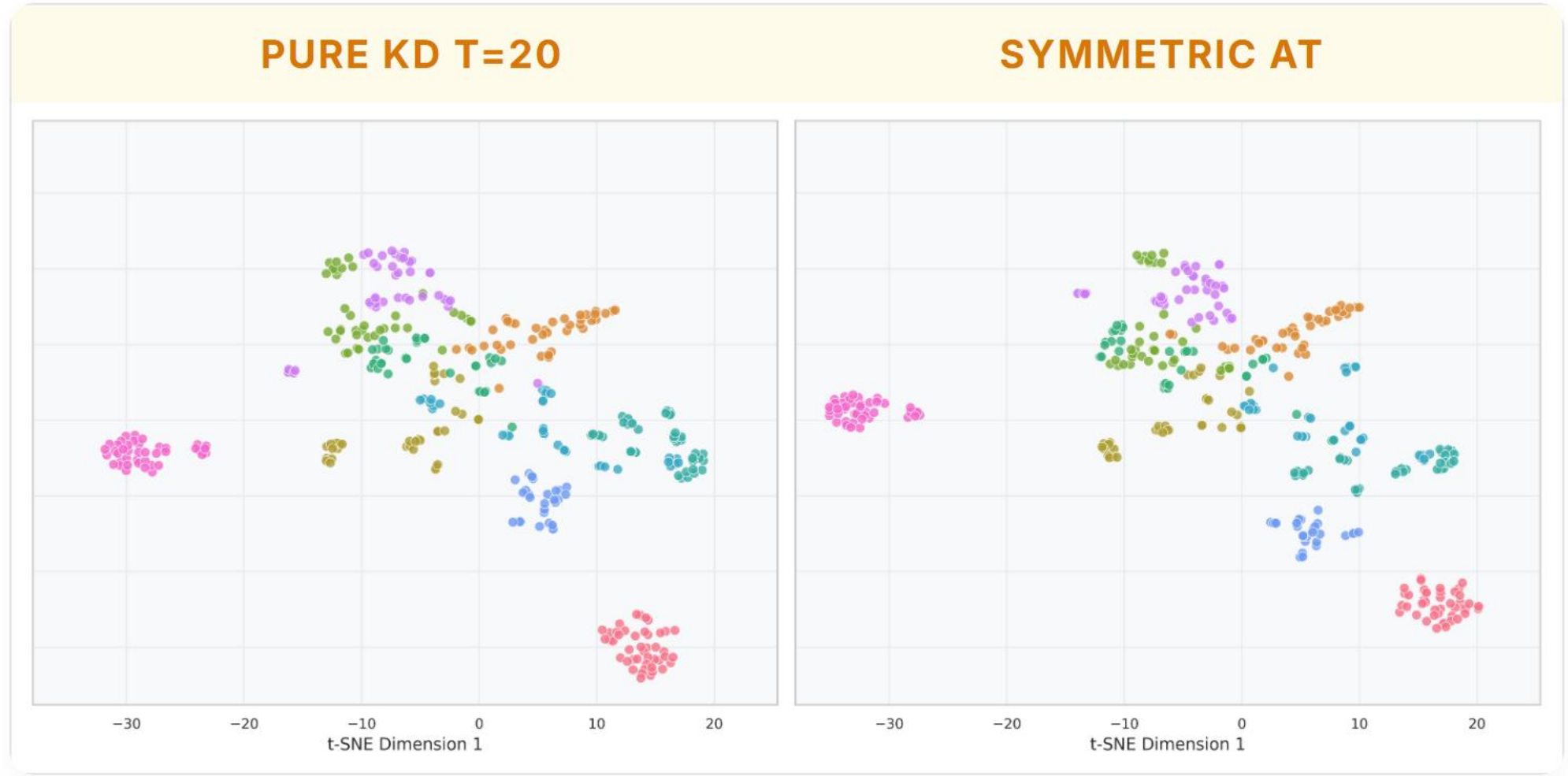
2D projections of pre-classifier embeddings — 10 selected classes



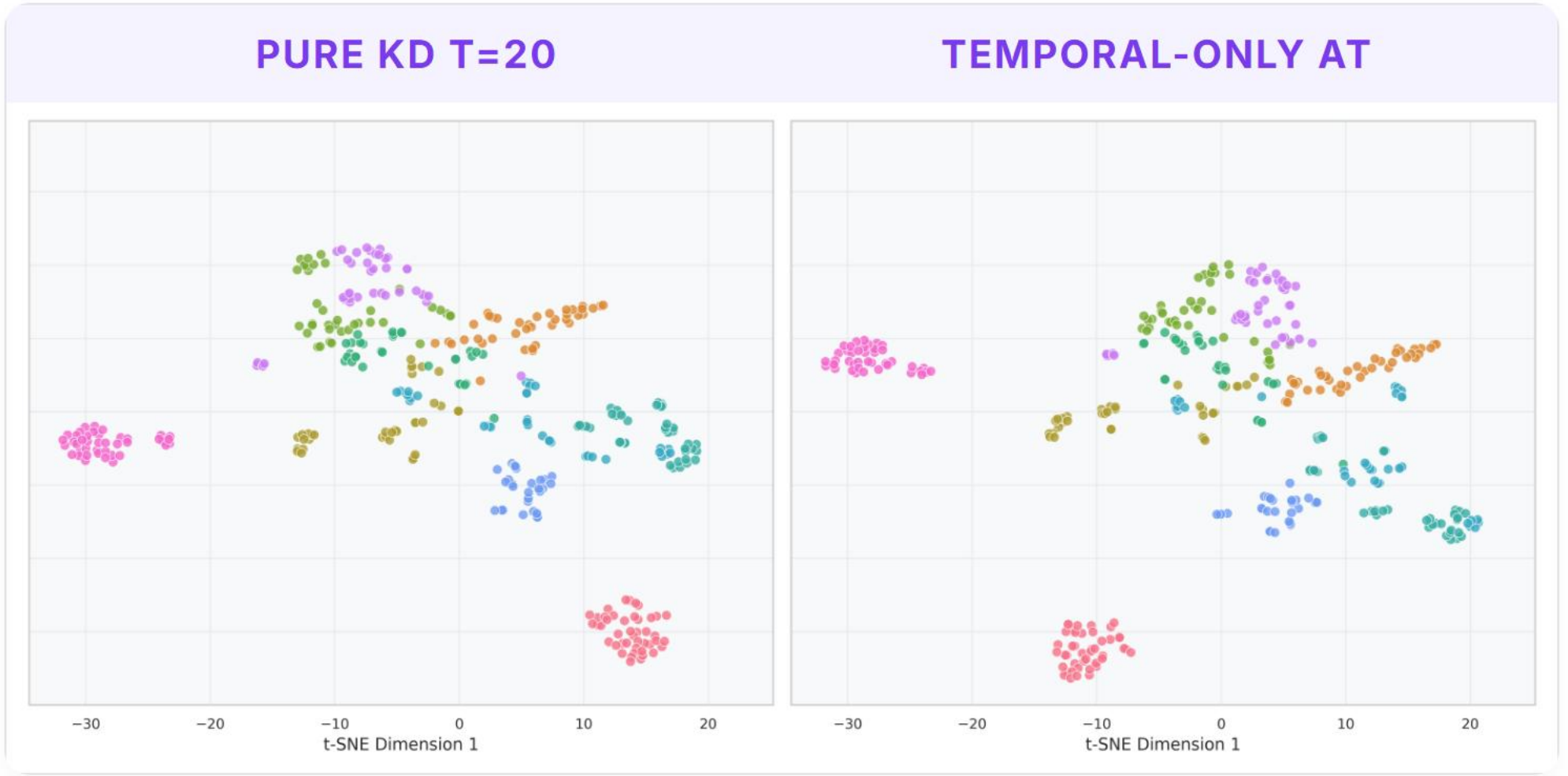
The distilled model (T=20) shows significantly tighter and more distinct class clusters compared to the baseline student, successfully inheriting the well-separated semantic boundaries of the teacher model.

Attention Transfer (AT)

Capacity gap impact on intermediate feature alignment



Forced spatial alignment acts as noise due to Student's depthwise convolution capacity limit.



Removing spatial constraints ($\beta_s=0$) attenuates regression, as temporal dynamics are more transferable.

Born-Again Networks & Knowledge Collapse

Iterative self-distillation MobileNet3D → MobileNet3D

$\alpha = 0.5, T = 2.5$

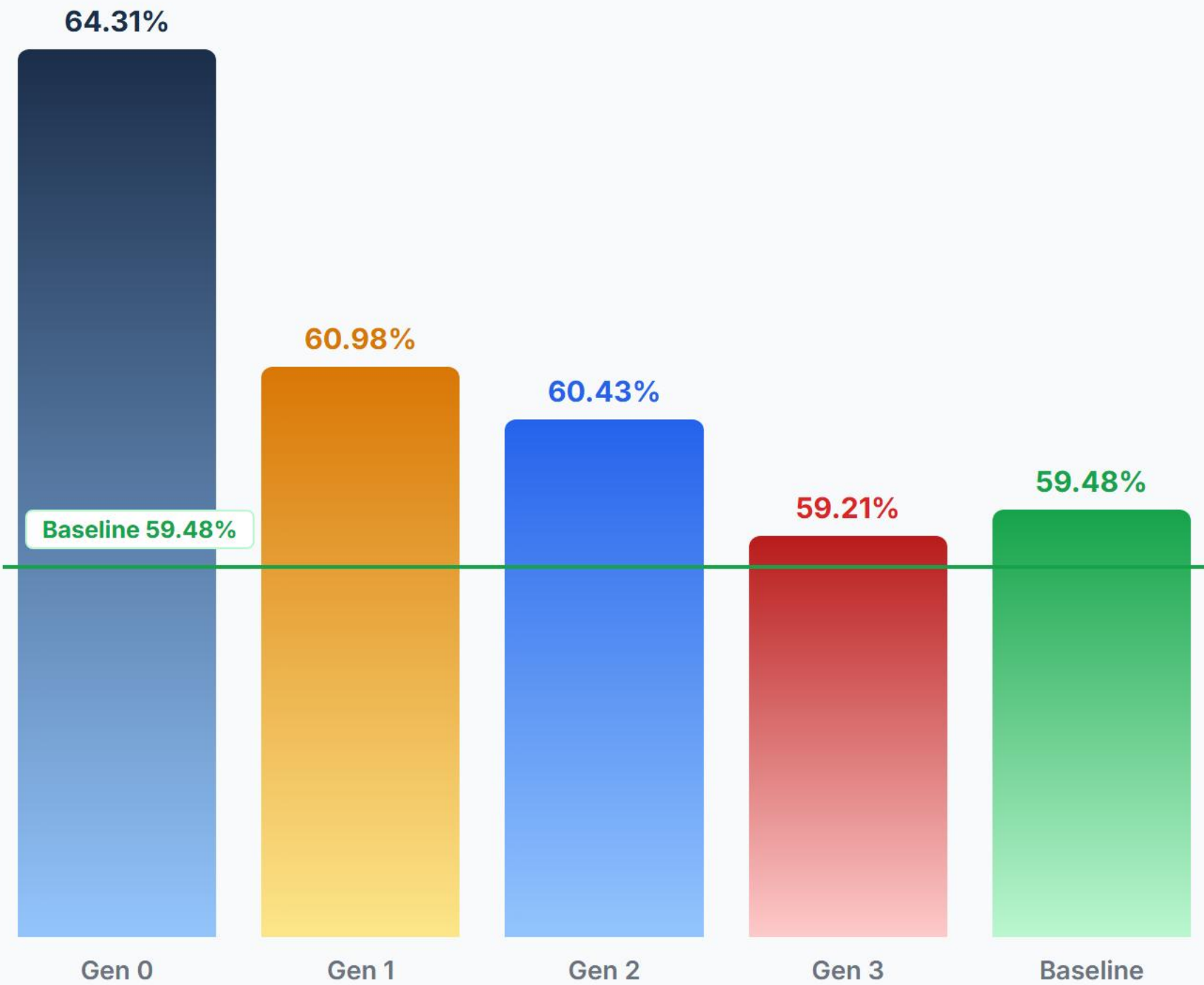
Iterative Self-Distillation

GEN	TEACHER	T	A	ACCURACY
Gen 0	3D ResNet-50	8.0	0.7	64.31%
Gen 1	MobileNet3D Gen 0	2.5	0.5	60.98%
Gen 2	MobileNet3D Gen 1	2.5	0.5	60.43%
Gen 3	MobileNet3D Gen 2	2.5	0.5	59.21%
Baseline	No Teacher (Scratch)	—	—	59.48%

Considerations

A compressed student (64.31%) lacks the visual representational capacity to act as an effective teacher. Distilling at **T=2.5** fails to prevent error propagation, causing progressive degradation back to baseline (59.48%).

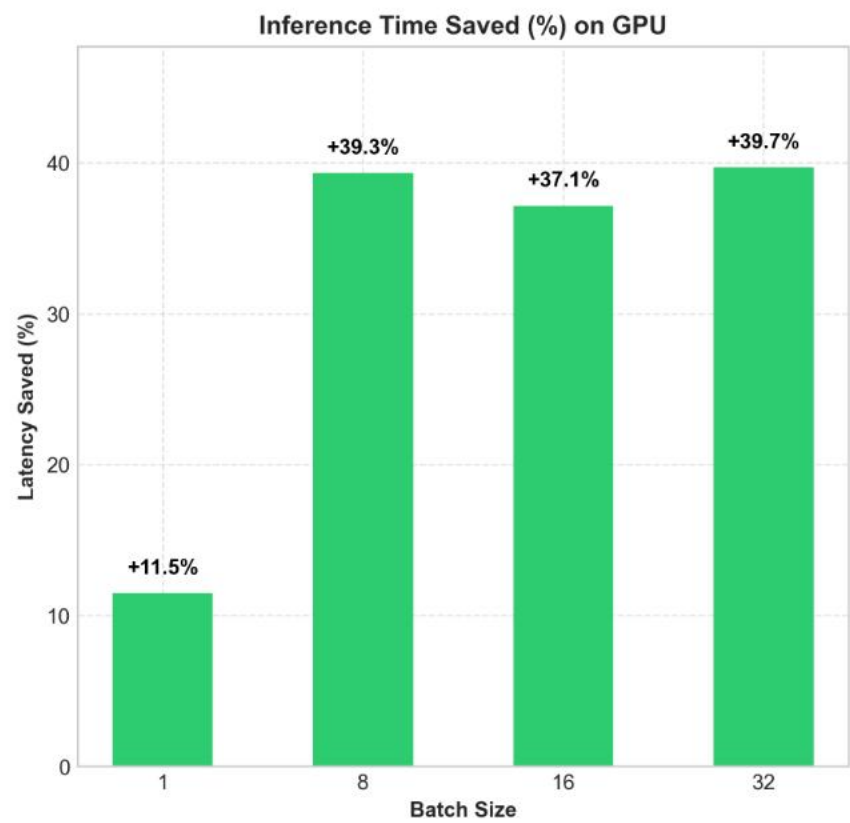
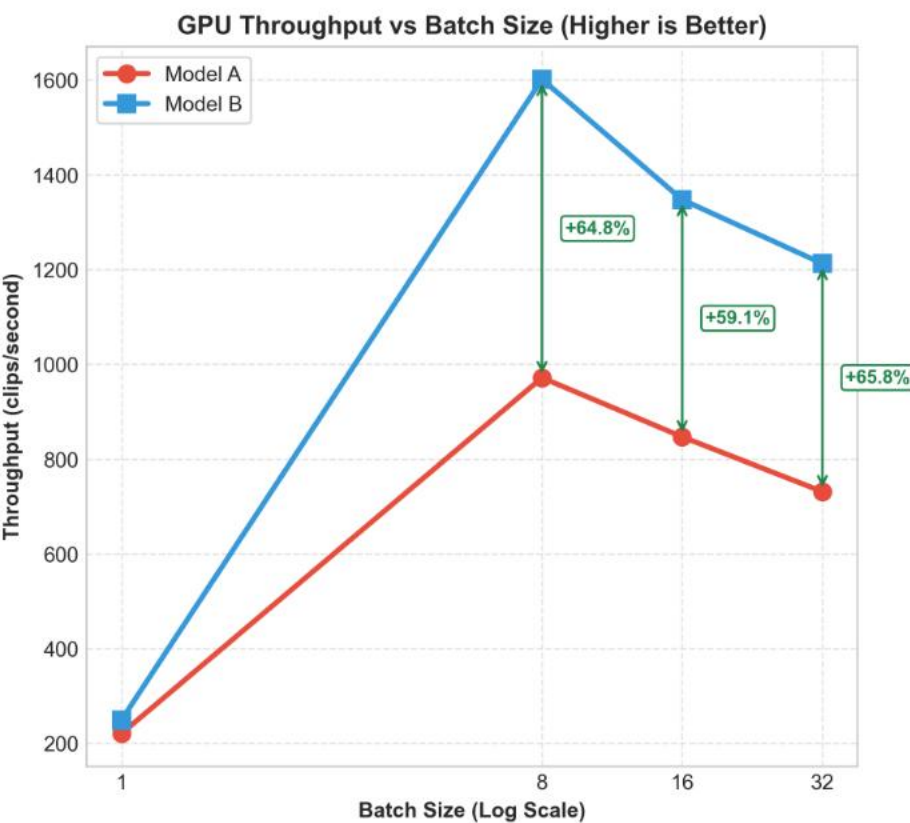
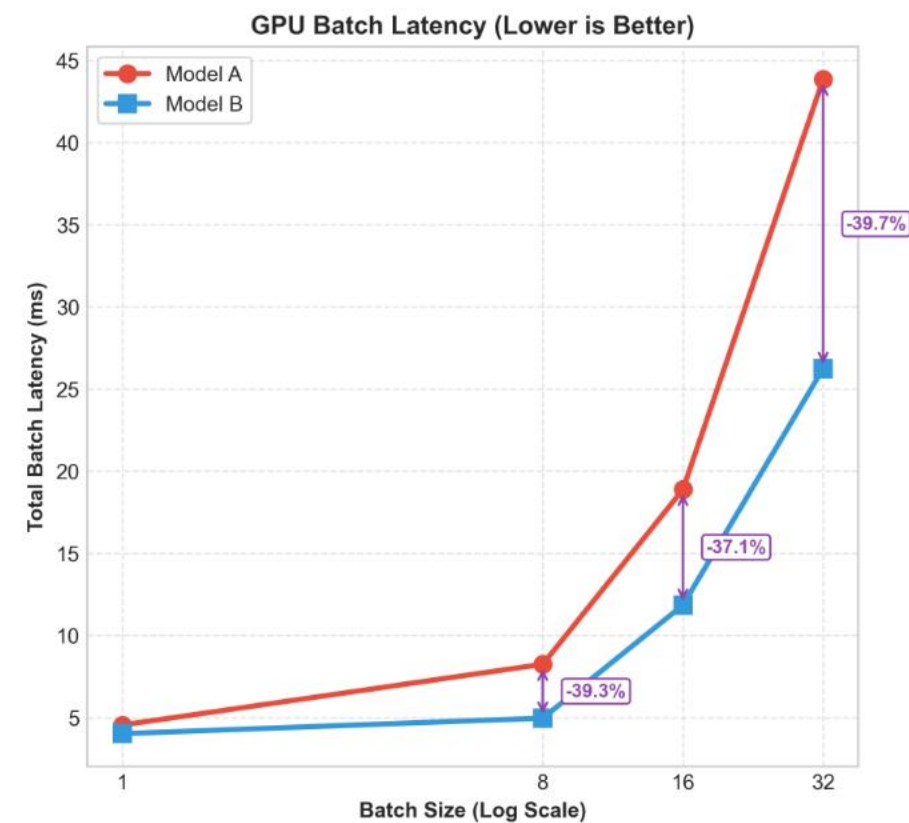
ACCURACY ACROSS GENERATIONS



Cross-Frame Distillation

Asymmetric temporal sub-sampling (T=20, a=0.7) — 16f Student vs 24f Teacher

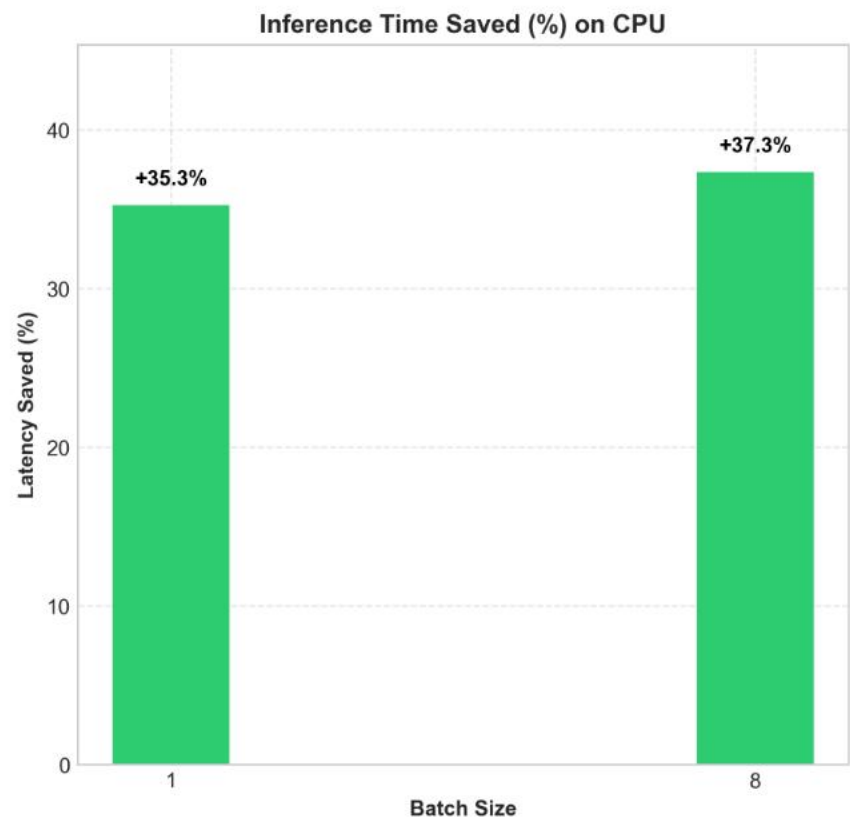
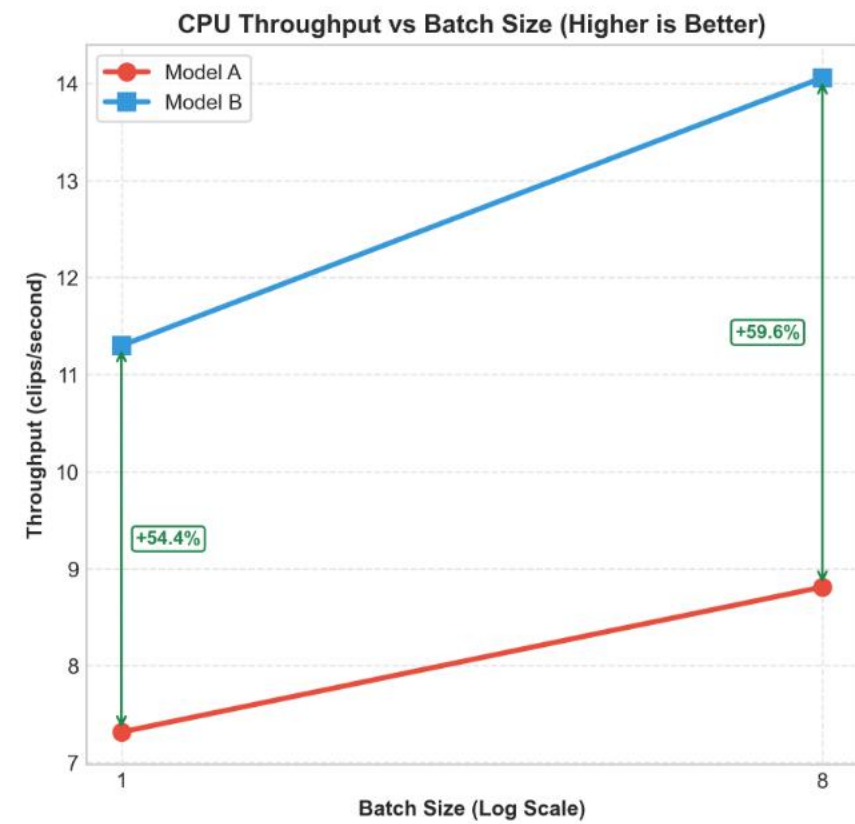
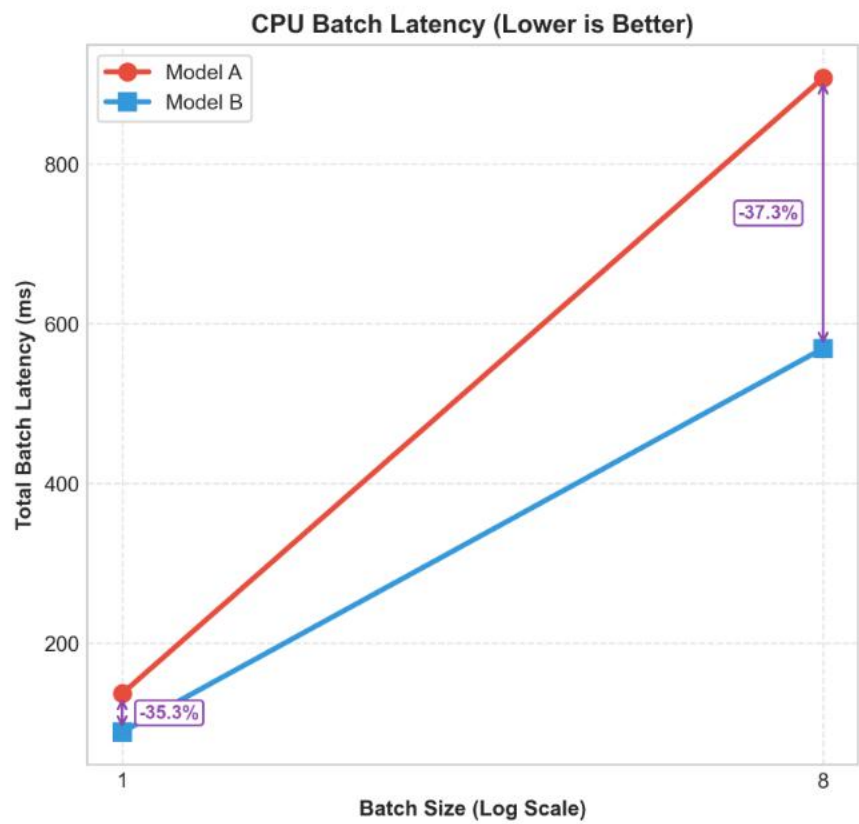
Test set inference Benchmark (GPU & CPU): Model A (24f) (Top-1: 65.85%) vs Cross Frame Model B (16f) (Top-1: 62.68%)



PERFORMANCE GAP
-3.17%

65.85% (24f) → 62.68% (16f)

Minimal accuracy cost for temporal compression



INFERENCE SPEEDUP

2.11x

CPU latency halved (-52.6%)

GPU throughput boosted by +68.8%

Key Takeaways & Future Work

NO FRAME DIST. (24F)

CROSS-FRAME DIST. (16F)

ACCURACY

65.85% +6.37 pp

vs. 59.48% Baseline

ACCURACY

62.68% +3.20 pp

vs. 59.48% Baseline

CPU LATENCY

53.52 ms

24-frame computational load

CPU LATENCY

25.36 ms -52.6% (2.11x)

Latency halved on Edge CPU

GPU THROUGHPUT

991 v/s

Standard throughput

GPU THROUGHPUT

1673 v/s +68.8%

Register & Tensor Core alignment

Lessons Learned

The spatial capacity gap limits intermediate feature transfer. Logit distillation at high temperature (\$T=20\$) is the most stable guide.

Key Limitations

- (KD) High temperature (T=20) over-smooths fast, cyclic actions.
- (AT) Spatial capacity gap limits intermediate Feature Distillation.
- (BAN) Error propagation & bias accumulation in Born-Again Networks.

Future Work

- **Variable Temporal Stride:** Train with dynamic frame sampling rates to prevent high-T over-smoothing on fast, cyclic actions.
- **Flexible Feature Alignment:** Introduce temporary learnable adapter blocks (e.g. 1x1 convs) during training to bridge the AT capacity gap.

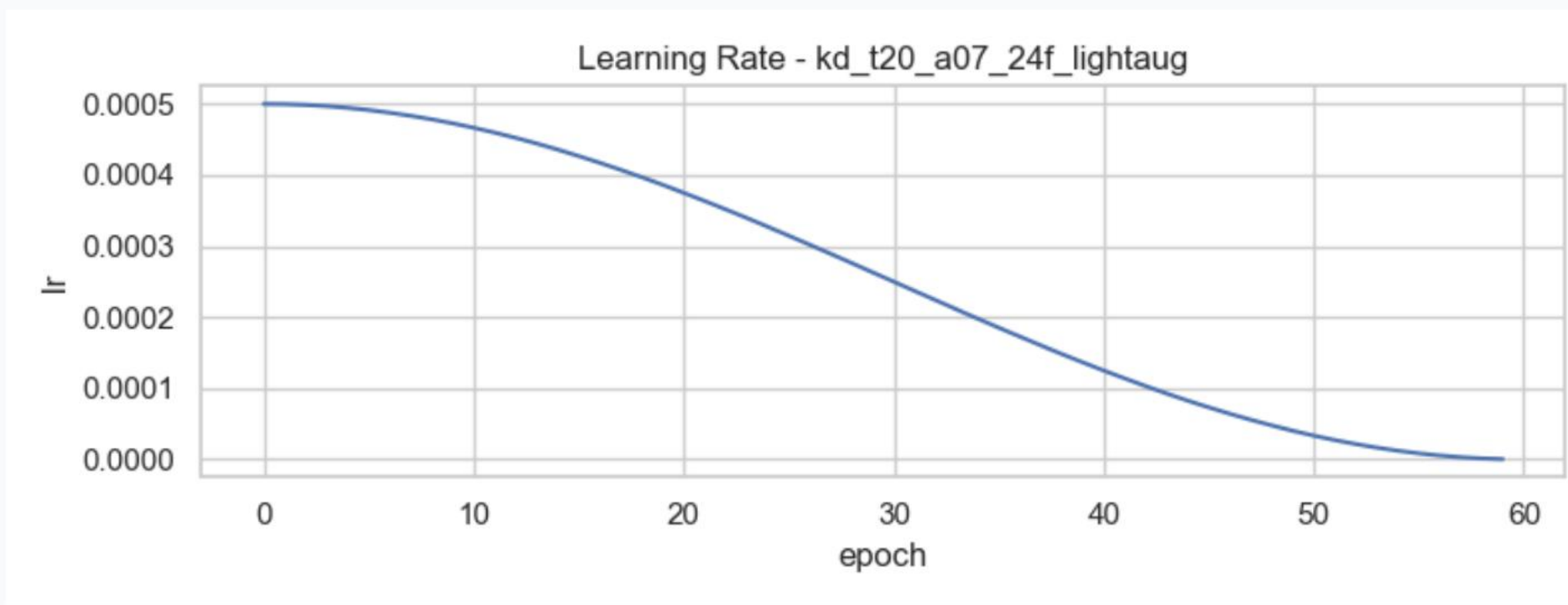
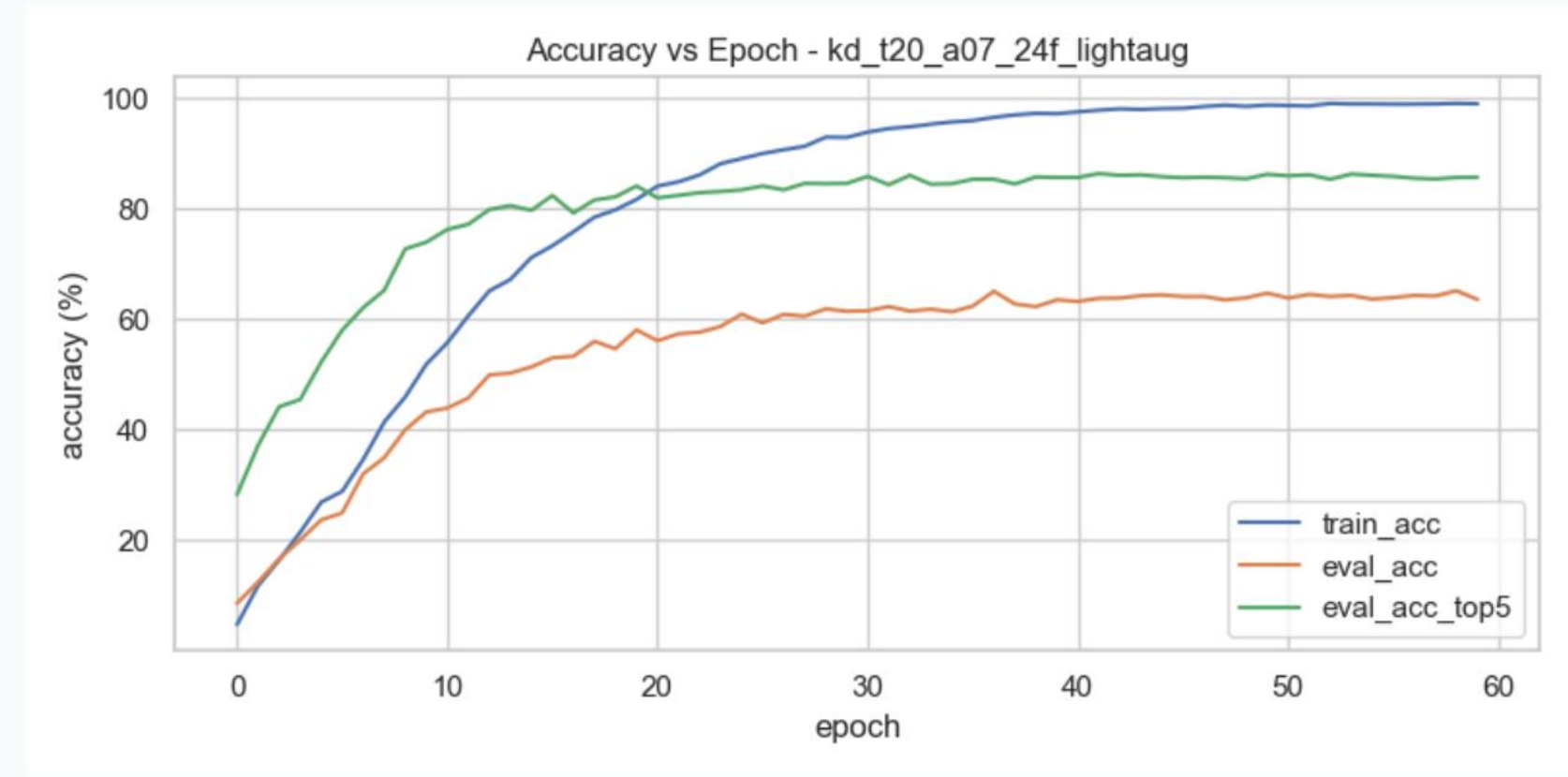
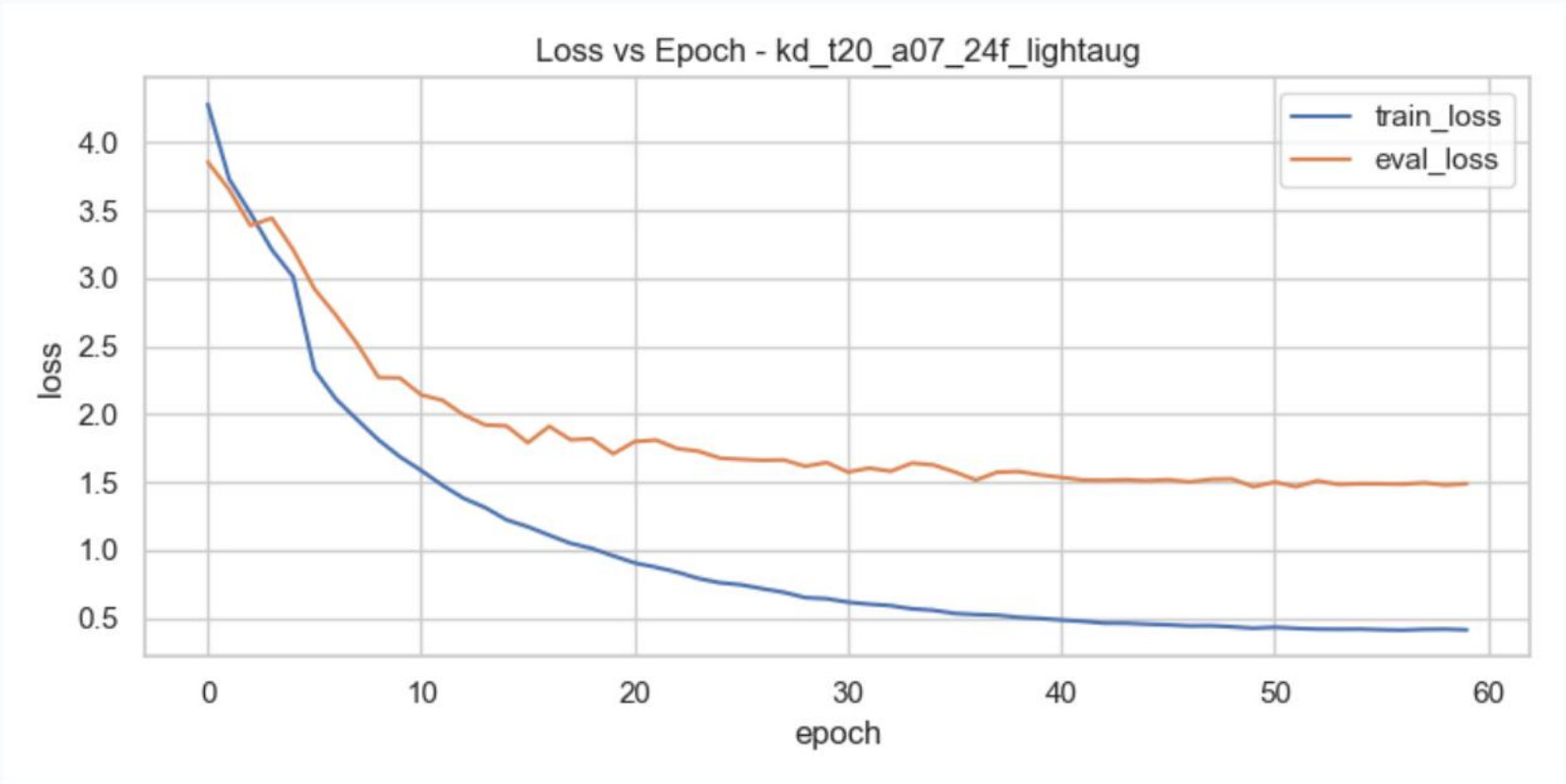
ADDITIONAL MATERIAL

Backup Slides

Quantitative plots, qualitative detailed analyses, and CM predictions.

Training Curves — KD T=20, a=0.7

Loss, accuracy, and LR schedule over 60 epochs



Stability & Convergence

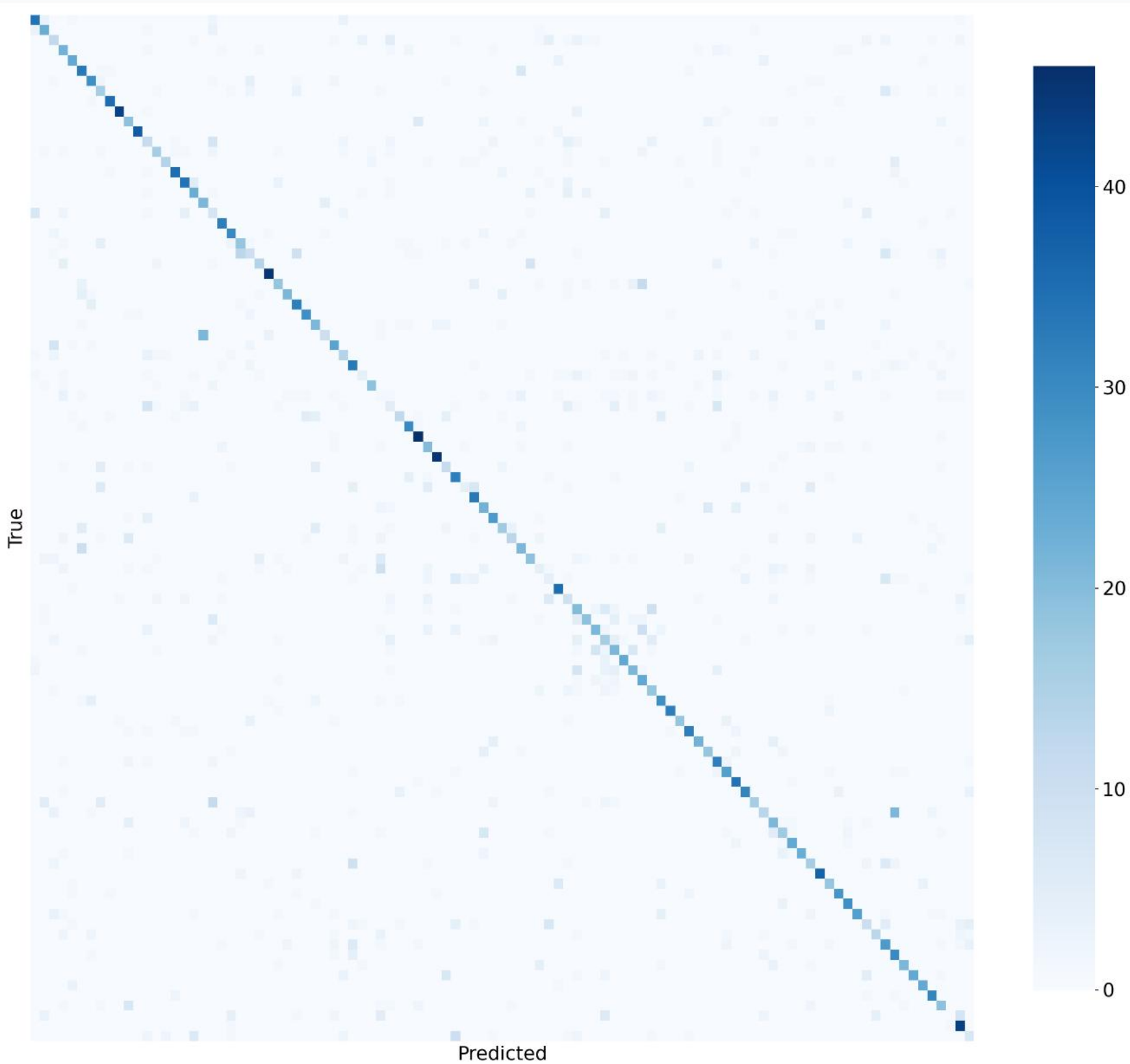
Stable training over 60 epochs. First 10 epochs serve as warmup with hard targets before soft distillation loss activates.

No strong overfitting divergence — regularization works optimally with T=20.

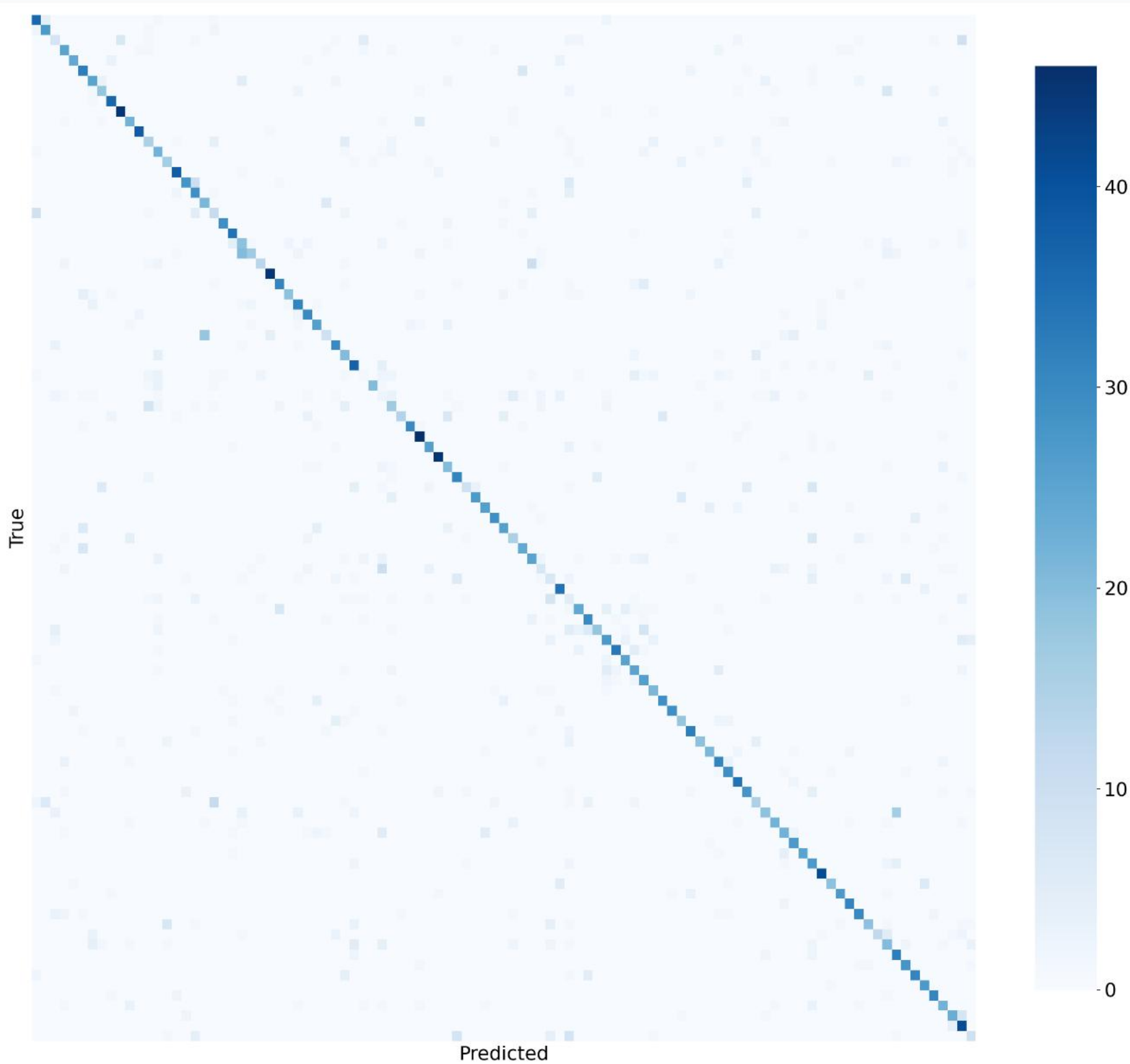
Confusion Matrix & Error Analysis

Baseline vs KD T=20 - Full normalized matrices

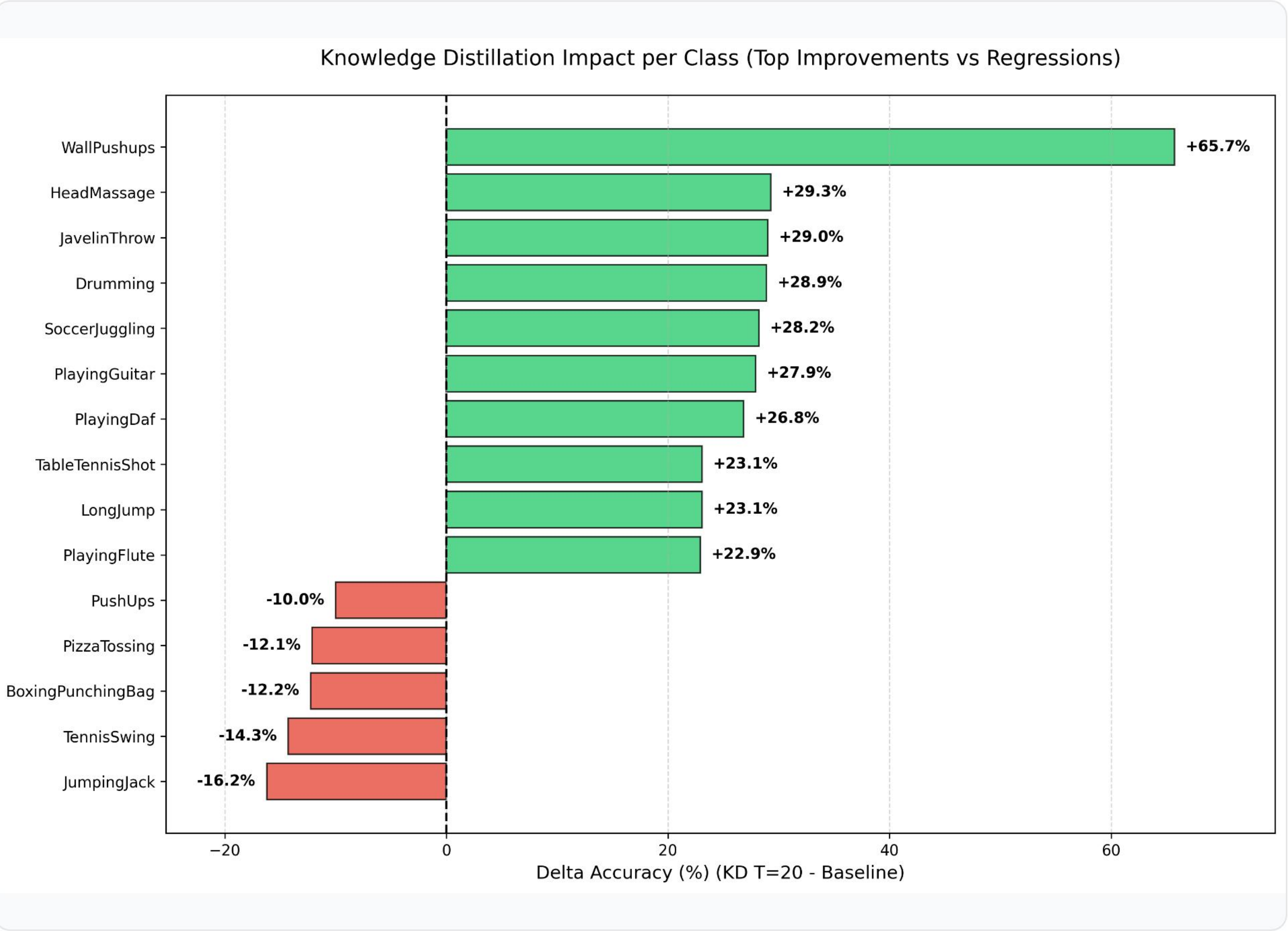
BASELINE - 59.48%



KD T=20 - 65.85%



Per-Class Accuracy: KD T=20 vs Baseline



Improvements

Spatially complex classes (e.g. **WallPushups**, **HeadMassage**) gain most from the Teacher's visual filters.

Regressions

Fast, cyclic actions (**JumpingJack**, **TennisSwing**) suffer from temporal over-smoothing at high T.

Shared Challenges

Classes with high structural ambiguity (e.g. **HandstandWalking**, **Nunchucks**) or strong mutual confusion (e.g. **MoppingFloor**) remain unresolved in both models.

Attention Transfer Loss: Before and After

Decomposing 3D video features to resolve the spatial-temporal collapse

Before: Single Collapsed Loss

$$\mathcal{L}_{AT} = \sum_l || \text{normalize}(\text{mean}_c(F^2))_{flat} - ... ||^2$$

- **Block Mismatch:** Hooked ResNet stages [3, 4, 5]. Stage 5 is the classifier head (produces 2D logits [B, 101]), which caused runtime crashes and zero gradient contribution.
- **1D Interpolation Mismatch:** Resizing a flattened 1D vector of shape [T × H × W] treats space and time as a single coordinate line. This mathematically mixed pixels across frame boundaries and unrelated spatial points, destroying feature geometry.
- **Temporal Signal Dilution:** Normalizing the entire flattened space meant temporal peaks were washed out by spatial dimensions.

59.14%

After: Spatial & Temporal Decomposition

$$\mathcal{L}_{AT} = \beta_s \mathcal{L}_{spatial} + \beta_t \mathcal{L}_{temporal}$$

- **Correct Block Pairing:** Hooked ResNet stages [2, 3, 4] (intermediate 3D features: [B, C, T, H, W]), correctly matching student stages [2, 4, 6].
- **2D Bilinear Interpolation:** Spatial attention is calculated as mean_{c,t}(F²) and resized spatially via 2D bilinear interpolation, preserving frame layout.
- **Temporal AT Isolation:** Temporal attention is calculated as mean_{c,h,w}(F²) yielding [B, T], which is transferred directly without any interpolation noise.

64.58% · 65.03%

Symmetric: β_s=0.05, β_t=0.05 · Temporal: β_s=0, β_t=0.10